Multi-Sensor Magnetometer PCB

Design, Operation, and Programming Reference

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October 19, 2026

Prepared for internal lab use. Intended audience: future lab members and primary user Chunya Wang.

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1 Introduction and Purpose

1.1 Background

This document describes the design, assembly, and operation of the Magnetometer PCB developed for Dr. Elliott's Coherent and Quantum Optics Lab at Purdue University. The board measures three-dimensional magnetic field vectors at multiple spatial positions using MEMSIC MMC5983MA AMR sensors, in support of Chunya Wang's experimental work in the lab.

The system uses two identical PCBs connected in series via a flat flexible cable (FFC). Each board carries four sensors multiplexed over a single I^2C bus, all controlled by a shared Arduino Nano Every microcontroller.

1.2 Design Goal

The core measurement objective is to estimate the magnetic field at the **center point** of the sensor array by averaging the readings from all four sensors on a board. Because the sensors are arranged symmetrically around a circular cutout — at the 12, 3, 6, and 9 o'clock positions — their mean approximates the field value at the geometric center, where the experimental object (optical cavity, glass cell, etc.) is located.

For each field component, the center-point estimate is:

$$\bar{B}_x = \frac{1}{N}\sum_{i=1}^{N} B_{x,i}, \quad \bar{B}_y = \frac{1}{N}\sum_{i=1}^{N} B_{y,i}, \quad \bar{B}_z = \frac{1}{N}\sum_{i=1}^{N} B_{z,i} \tag{1}$$

where $N = 4$ per board (or $N = 8$ for the two-board system). This averaged field is then monitored and actively minimized — using external field-cancellation coils or passive shielding — to reduce residual magnetic field at the experiment site and eliminate its impact on sensitive quantum optics measurements.

1.3 Scope of This Document

A future student or Chunya should be able to use this document alone to bring the system from unpowered to collecting averaged field data. It covers: circuit design and component selection, PCB layout and assembly, physical installation in the vacuum chamber setup, two-board configuration, firmware operation, and troubleshooting.

2 Circuit Design and Architecture

2.1 System Block Diagram

Figure 1 shows the full two-board system. The Arduino Nano Every acts as the I^2C master. Both boards share the same SDA, SCL, RST, +5 V, and +3.3 V lines routed through the FFC connector. Each board contains an LM3940 voltage regulator, one TCA9546A I^2C multiplexer, and four MMC5983MA sensors.

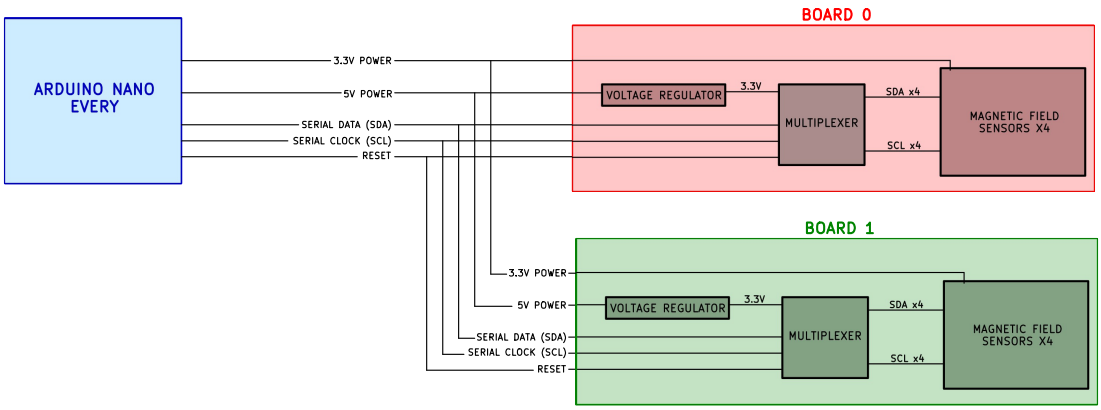


Figure 1: Full system block diagram. The Arduino Nano Every drives two identical sensor boards (Board 0 and Board 1) via a shared FFC bus.

2.2 Top-Level Schematic

Figure 2 shows the top-level KiCad schematic for one board. The FFC connector routes RST, SCL, SDA, and the four mux channel pairs (SC0/SD0 through SC3/SD3) to the TCA9546A sub-sheet and the four MMC sub-sheets. Two mounting holes (H1, H2) are also indicated.

Figure 2: Top-level schematic (one board) showing FFC connector, TCA9546A mux, and four MMC5983MA instances.

2.3 Component Summary

Table 1: Key Components (per board)

Component	**Part Number**	**Qty**	**Function**
Magnetic field sensor	MMC5983MA	4	3-axis AMR magnetometer
I^2C multiplexer	TCA9546AD	1	4-channel bus switch
Voltage regulator	LM3940IMPX	1	5 V $\rightarrow$ 3.3 V LDO
FFC connector (in/out)	*[PN]*	2	Board-to-board bus
Bulk output cap	33 µF tantalum electrolytic	1	LDO output stabilization
Mux bypass cap	10 µF ceramic (X5R/X7R)	1	TCA9546A VCC
Sensor VDD bypass cap	1 µF ceramic (X5R/X7R)	4	MMC5983MA VDD
Sensor CAP bypass cap	10 µF ceramic (X5R/X7R)	4	MMC5983MA CAP pin
I^2C pull-up resistors	2.7 kΩ	11	All SDA/SCL lines

2.4 MMC5983MA Magnetic Field Sensor

The MMC5983MA (MEMSIC) is a 3-axis anisotropic magnetoresistance (AMR) magnetic field sensor in a $2.0 \times 2.0 \times 1.0$ mm LGA-16 package. It measures B_x, B_y, and B_z simultaneously and communicates over I^2C at a fixed address of `0x30`. Key specifications are in Table 2.

Table 2: MMC5983MA Key Specifications [1]

Parameter	**Value**	**Notes**
Supply voltage (VDD)	1.71–3.6 V	
Full-scale range	±8 G	
Resolution (18-bit mode)	0.0625 mG/LSB	18-bit output
RMS noise	0.4 mG (typ.)	Continuous mode
Output data rate	up to 1000 Hz	
I^2C address	`0x30` (fixed)	
Package	LGA-16, 2.0×2.0 mm	

2.4.1 Interface Selection: I^2C vs. SPI

The MMC5983MA supports both I^2C and SPI interfaces, selected by the state of the SPL_CS pin at power-up. Table 3 compares the two options.

Table 3: MMC5983MA Interface Comparison

Property	**I²C**	**SPI**
Bus lines required	2 (SDA, SCL) shared	4 per sensor (MOSI, MISO, SCK, CS)
Max clock	400 kHz (Fast Mode)	10 MHz
Multiple sensors	Requires address mux	Requires individual CS lines
Wiring complexity	Low	High (4 traces per sensor)
Throughput bottleneck	2 ms conversion time	2 ms conversion time
Mux compatibility	TCA9546A works directly	No equivalent mux available
SPL_CS pin	Float or leave NC	Drive low to enable

I²C was chosen for this design because the per-sensor throughput is limited by the 2 ms measurement conversion time — not bus speed — making the 10 MHz SPI clock advantage irrelevant. I²C requires only two shared bus traces regardless of sensor count, and integrates naturally with the TCA9546A multiplexer (Section 2.5). SPI would require four dedicated traces per sensor (36 traces for 8 sensors plus CS decode logic), dramatically increasing PCB complexity with no meaningful speed benefit. SPL_CS is left unconnected (floating high at power-up), which selects I²C mode per the MMC5983MA datasheet.

2.4.2 Pin Configuration

Table 4: MMC5983MA Pin Assignment (this design)

Pin	**Name**	**Connected to**	**Notes**
1	SCL/SPL_SCK	SCx (from mux channel)	I²C clock
2	VDD	+3.3 V via 1 µF ceramic	Supply; 1 µF MLCC adjacent to pin
5	SPL_CS	NC (floating)	Selects I²C mode at power-up
6	SPL_SDO	NC	SPI not used
9	GND	GND	
10	CAP	GND via 10 µF ceramic	Charge pump for SET/RESET coil; value required by datasheet
13	VDDIO	+3.3 V	I/O voltage
15	INT	NC	Interrupt not used; firmware polls
16	SDA/SPL_SDI	SDx (from mux channel)	I²C data

2.4.3 SET/RESET Coil

The MMC5983MA contains an on-chip coil for re-magnetizing the Permalloy sensing element. Two operations are available:

SET (write `0x08` to `CONTROL_0`): magnetizes the film in the positive direction, establishing a known reference and removing any offset accumulated from strong external field exposure.

RESET (write `0x10` to `CONTROL_0`): magnetizes the film in the opposite direction. Taking the difference of a SET and RESET measurement:

$$B = \frac{B_{\text{SET}} - B_{\text{RESET}}}{2} \tag{2}$$

cancels the temperature-dependent bridge offset and substantially improves measurement accuracy. Wait at least 1 µs after each pulse before triggering a measurement.

The 10 µF ceramic capacitor on the CAP pin is the energy reservoir for the SET/RESET coil's internal charge pump. It is required by the datasheet and must not be omitted or undersized — doing so will weaken the pulse and degrade offset cancellation.

2.4.4 Fixed I²C Address and the Multiplexer Requirement

All MMC5983MA devices share the fixed address `0x30`. Placing more than one sensor on a single bus causes address collisions. The TCA9546A multiplexer resolves this by isolating each sensor on its own switched bus segment, presenting only one sensor to the master at a time.

2.5 TCA9546A I²C Multiplexer

The TCA9546AD (Texas Instruments) is a low-voltage 4-channel I²C bus switch that sits between the Arduino (master) and the four MMC5983MA sensors. Its address is set by pins A0–A2: both boards use A2 = A1 = GND; A0 differentiates the boards (Table 5).

Table 5: TCA9546A Address Strapping per Board

Board	A2	A1	A0	I²C Address
Board 0	GND	GND	GND	`0x70`
Board 1	GND	GND	VCC	`0x71`

The downstream channel is selected by writing a single byte to the mux: bit n enables channel n. Writing `0x00` disconnects all channels (recommended after each sensor read to prevent bus contention).

Table 6: TCA9546A Channel Select Byte

Channel	Byte (hex)	Sensor
0	`0x01`	U4 / MMC_0
1	`0x02`	U3 / MMC_1
2	`0x04`	U5 / MMC_2
3	`0x08`	U6 / MMC_3
None	`0x00`	All disconnected

Each I^2C segment (upstream and each of the four downstream channels) has its own 2.7 kΩ pull-up pair to +3.3 V (11 resistors total, R1–R11). Pull-ups are required on each downstream segment because the TCA9546A is a switch, not a buffer: disconnected segments have no pull-ups unless they are present locally. A 10 µF ceramic bypass cap (C1) decouples the mux VCC pin.

The active-low `/RESET` pin is connected to the RST net from the FFC connector, allowing the Arduino to perform a hardware bus reset without power-cycling the board.

2.6 LM3940IMPX Voltage Regulator

The LM3940IMPX (U2) converts the 5 V Arduino USB rail to 3.3 V for all sensors and the mux. A **33 µF tantalum electrolytic capacitor** (C2) is required on the output. This value and type are specified in the LM3940 datasheet: the device requires ≥33 µF with ESR in the range 0.3–22 Ω for loop stability. Ceramic MLCC capacitors have ESR below 0.01 Ω and will cause the LDO to oscillate; tantalum electrolytic provides the correct ESR naturally. **Do not power sensor VDD from the Arduino 5 V pin directly.**

2.7 FFC Connector and Daisy-Chain Bus

Each board has two FFC connectors (J4 PIN_IN, J1 PIN_OUT) for daisy-chain expansion. The 16-pin connector carries GND, +5 V, +3.3 V, SCL, SDA, and RST. The regulated +3.3 V is forwarded through PIN_OUT to power downstream boards directly.

Cable orientation: Use a Type-B FFC cable (Top-on-One-Side, Bottom-on-the-Other). Inserting a Type-A cable will reverse the pinout and short power to ground. Verify power and signal alignment before applying power.

2.8 Circuit Schematics

2.8.1 Power Distribution

Figure 3 shows the power and connector schematic. The LM3940IMPX (U2) takes +5 V from the FFC input (J4) and outputs the +3.3 V (VCC) net stabilized by C2. Both supply voltages and all signal lines pass through to J1 (PIN_OUT) for the next board in the chain.

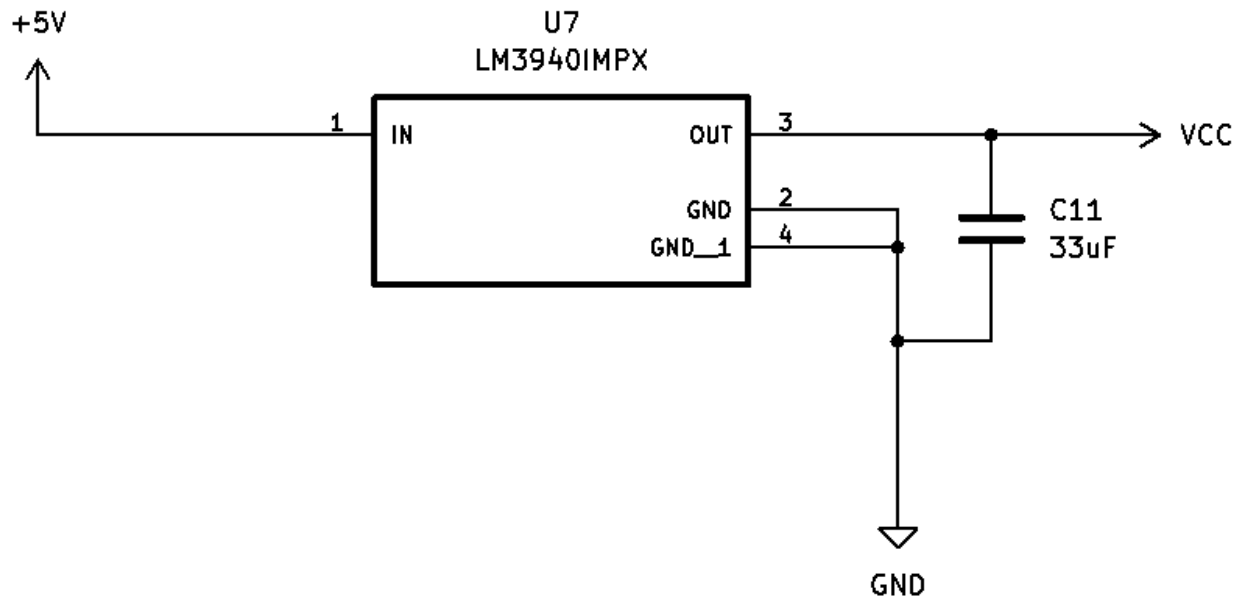


Figure 3: Power distribution schematic. U2 (LM3940IMPX) converts +5 V to +3.3 V (VCC). J4 is the input FFC connector; J1 passes all signals downstream.

2.8.2 I²C Multiplexer

Figure 4 shows the TCA9546A schematic. Upstream SDA and SCL enter pins 15 and 14 with 2.7 kΩ upstream pull-ups (R1–R3). Each downstream channel pair (SD0/SC0 through SD3/SC3) has its own 2.7 kΩ pair (R4–R11). A0 (pin 1) is the board-address strap; RESET (pin 3) connects to the RST net.

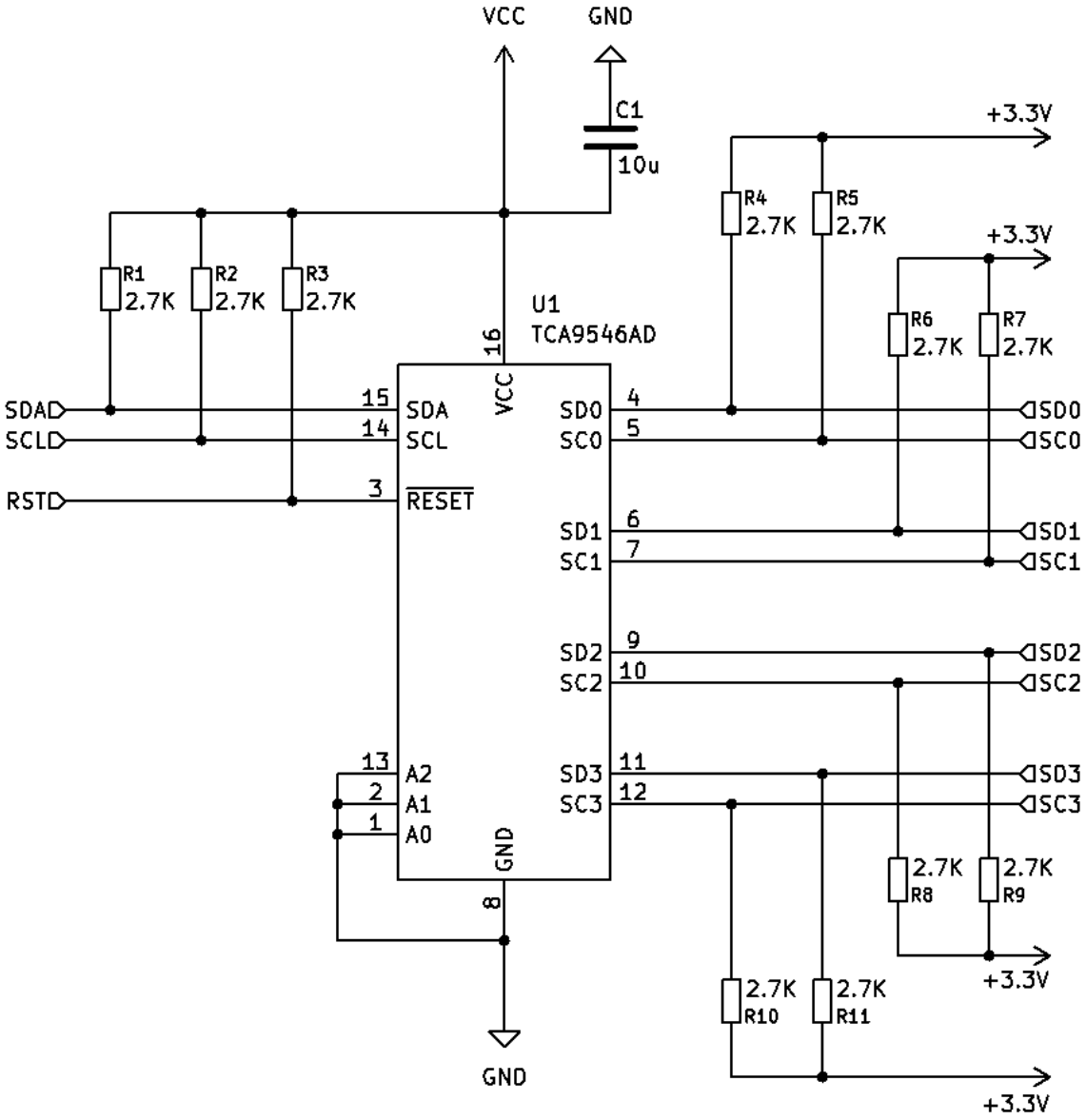


Figure 4: TCA9546AD multiplexer schematic. All pull-ups are 2.7 kΩ to +3.3 V. A0 selects the board I²C address (Table 5).

2.8.3 MMC5983MA Sensor Circuit

All four sensors per board use the identical circuit; one representative instance is shown in Figure 5. Each sensor connects its SCx/SDx lines to a dedicated TCA9546A downstream channel. A 1 µF ceramic cap decouples VDD and a 10 µF ceramic cap connects the CAP pin to ground. The INT pin is left unconnected; SPL_CS and SPL_SDO are left unconnected to force I²C mode at power-up (Section 3).

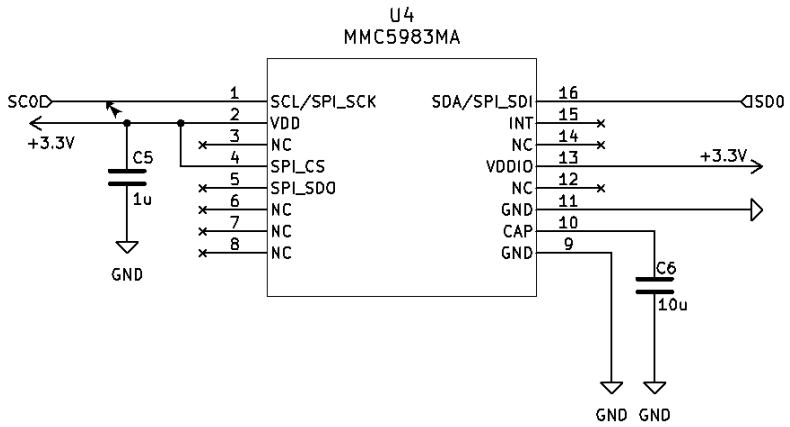


Figure 5: Representative MMC5983MA sensor circuit (U4, channel 0). All four sensor instances (U3–U6) are electrically identical; only the mux channel they connect to differs.

3 PCB Layout and Assembly

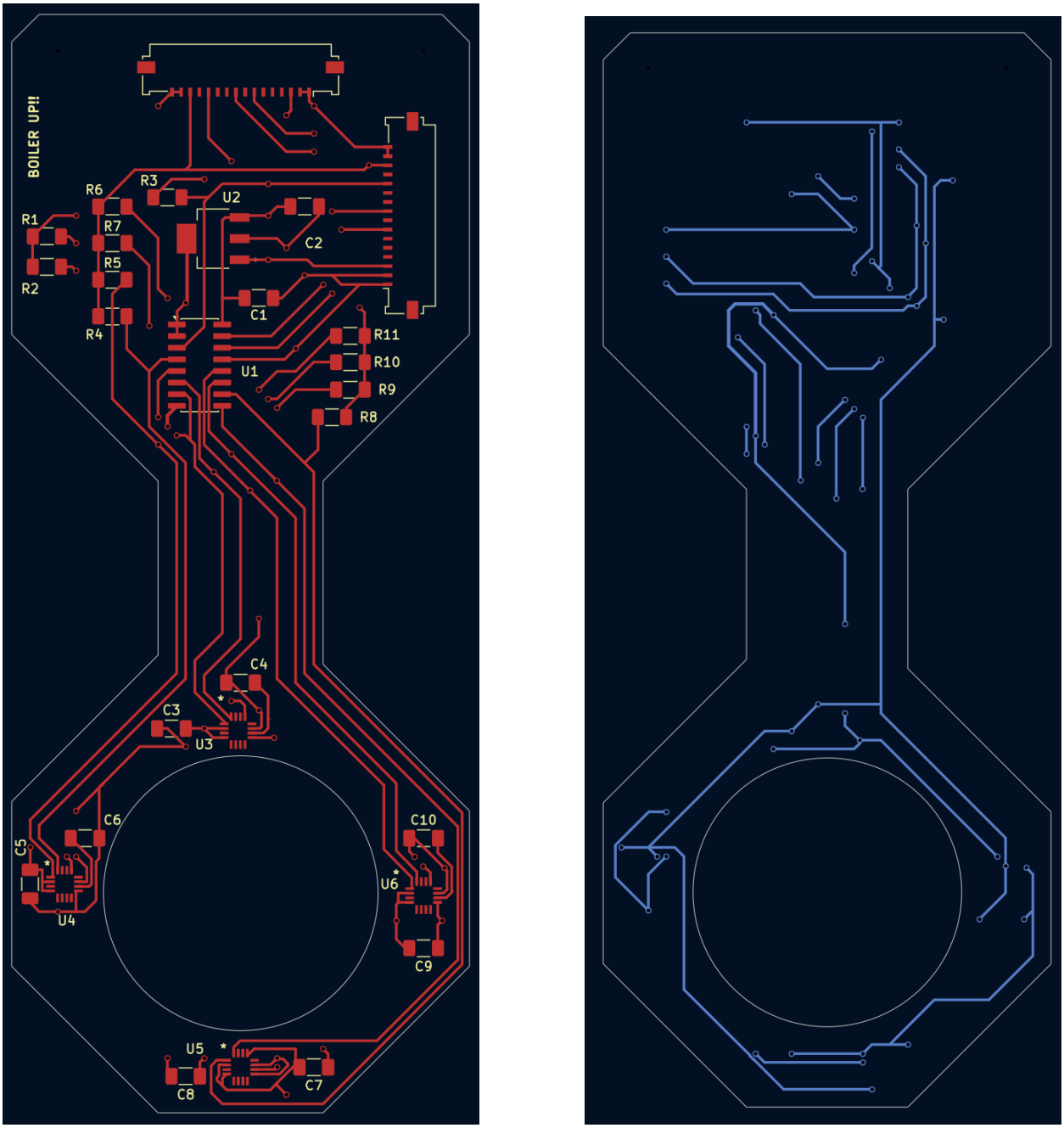
3.1 Board Overview

The PCB measures **50.05 mm × 120.05 mm** on a 2-layer, 1.6 mm FR4 substrate designed in KiCad 10.0.1. Fabrication specifications are listed in Table 7.

Table 7: PCB Fabrication Specifications

Parameter	**Value**
Board dimensions	50.05 mm × 120.05 mm
Layer count	2 (F_Cu top, B_Cu bottom)
Board thickness	1.6 mm
Min trace width	0.3 mm
Pad-to-pad clearance	0.2 mm

3.2 Top and Bottom Copper Layers



(a) Top copper (F_Cu) with silkscreen.

(b) Bottom copper (B_Cu).

Figure 6: PCB layout rendered from Gerber files. Top layer carries all active routing; the bottom layer provides return paths.

3.3 Board Shape and Functional Regions

The board has a distinctive hourglass profile divided into three functional regions:

1. **Control section (top octagon):** Houses both FFC connectors, the LM3940IMPX (U2) with its 33 µF tantalum output cap (C2), the TCA9546A (U1) with its 10 µF mux bypass cap (C1), and all 11 pull-up resistors (R1–R11). Two mounting holes are located at the top corners.
2. **Routing neck (center):** A narrow bridge that carries the eight I^2C channel traces (SD0–SD3, SC0–SC3) from the mux to the sensor section. Confining all routing to the neck keeps copper away from the magnetically sensitive sensor region.
3. **Sensor section (bottom octagon):** Contains the four MMC5983MA sensors (U3–U6) and their bypass capacitors (C3–C10), arranged around a **circular cutout** (approximately 30 mm radius). The cutout allows the board to be mounted concentrically around a cylindrical experimental object such as an optical cavity or glass cell.

3.4 Sensor Placement Around the Circular Cutout

The four sensors are positioned at the cardinal points of the cutout so that their spatial average approximates the field at the geometric center (see Equation 1). Clock positions are as viewed from the top (chip/component) side of the board; left and right are mirrored when viewed from the solder side.

Table 8: Sensor Positions Around Circular Cutout

Ref.	Channel	Clock position (chip side)	Notes
U3	1 (SC1/SD1)	12 o'clock (top)	
U4	0 (SC0/SD0)	9 o'clock (left)	
U6	3 (SC3/SD3)	3 o'clock (right)	
U5	2 (SC2/SD2)	6 o'clock (bottom)	

3.5 Magnetic Noise Minimization

- **Functional isolation:** All components carrying significant current (regulator, mux, connectors) are confined to the control section, physically separated from the sensors by the narrow neck.
- **Bypass cap proximity:** Each sensor's 1 µF VDD cap and 10 µF CAP pin cap are placed immediately adjacent to the sensor pads, minimizing high-frequency current loop area.
- **No ferromagnetics near sensors:** All passives in the sensor section are ceramic capacitors; no inductors or ferrite-core components are placed near the sensors.
- **Trace bundling:** I^2C channel traces are bundled through the neck with SDA/SCL pairs adjacent to reduce differential loop area.
- **Mounting hardware:** Use non-magnetic hardware (brass or nylon standoffs and screws). Stainless steel fasteners near the sensors will perturb readings.

3.6 Bill of Materials

Table 9: Bill of Materials (one board)

Ref.	Value / PN	Qty	Function	Supplier
U1	TCA9546AD	1	I^2C mux	TI / Digikey
U2	LM3940IMPX	1	5 V → 3.3 V LDO	TI / Digikey
U3–U6	MMC5983MA	4	3-axis sensor	MEMSIC / Digikey
J1, J4	FFC 16-pin	2	Board connector	Digikey
C1	10 µF ceramic	1	TCA VCC bypass	Digikey
C2	33 µF tantalum	1	LDO output cap	Digikey
C3,C5,C7,C9	1 µF ceramic	4	Sensor VDD bypass	Digikey
C4,C6,C8,C10	10 µF ceramic	4	Sensor CAP pin	Digikey
R1–R11	2.7 kΩ	11	I^2C pull-ups	Digikey
H1, H2	Mounting hole	2	Mechanical	—

3.7 Soldering Notes

- The MMC5983MA and TCA9546AD are LGA/VSSOP packages requiring reflow soldering. Hand-soldering is possible for the VSSOP but not recommended for the LGA sensors.
- Apply flux generously under LGA pads before placing components.
- After reflow, inspect all connections with a microscope or loupe. **Tombstoning** — a reflow defect where one end of an SMD component lifts off its pad as solder melts, leaving the part standing vertically on one end like a headstone — will cause I^2C failures that are difficult to diagnose electrically without close visual inspection.
- The LM3940IMPX is in a TO-263-4 SMD package; solder last to avoid heat damage to nearby sensors.

3.8 Initial Power-On Checklist

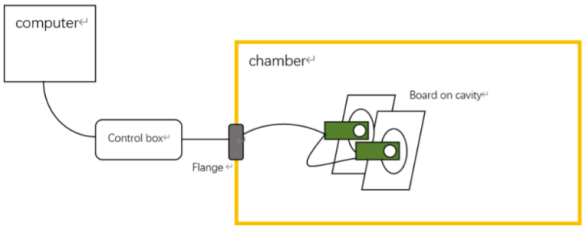
1. **Before connecting the Arduino:** Measure the +3.3 V output of U2 with a DMM. It should read 3.3 ± 0.1 V. If it reads 0 V or 5 V, check U2 orientation and C2 polarity.
2. Verify SDA and SCL read ≈3.3 V at rest (confirms pull-ups are present).
3. Connect the Arduino and upload the I^2C scanner sketch (Appendix D).
4. Confirm `0x70` (Board 0 mux) in the scan. For two boards, also confirm `0x71` (Board 1 mux).
5. Select each mux channel in turn and verify `0x30` appears on all channels (8 hits expected for two boards).

6. Run the test firmware and confirm plausible field readings ($\approx$0.2–0.6 G ambient Earth field).

4 System Integration and Physical Mounting

4.1 Overview

In the experimental setup, the magnetometer boards are mounted directly on the optical cavity inside the vacuum chamber, while the Arduino control box remains outside. The USB-serial cable exits the chamber through a feedthrough and connects to the data acquisition computer. Figure 7 illustrates the arrangement.



(a) System-level arrangement: sensor boards inside the vacuum chamber, Arduino control box outside.



(b) Actual experimental setup (earlier board revision shown; final board mounts identically). The PCB sits concentrically around the cylindrical cavity inside the aluminum frame. The FFC ribbon cable exits along the left edge; field-cancellation coil wiring is visible at the bottom.

Figure 7: Physical mounting arrangement inside the vacuum chamber.

4.2 Mounting the Board on the Cavity

The circular cutout in the sensor section of each PCB is designed to fit concentrically around the cylindrical optical cavity (or glass cell). To mount:

1. Slide the board over the cavity so the cylindrical object passes through the circular cutout. The four sensors will surround the cavity at 90° intervals.
2. Secure the board using the two mounting holes (H1, H2) in the control section with non-magnetic brass or nylon standoffs. **Do not use stainless steel or ferromagnetic hardware near the sensor section.**
3. **Sensor orientation — critical for two-board operation:** Both boards must be mounted with their chip sides facing the *same direction* relative to the cavity axis. Because each MMC5983MA sensor has a fixed internal axis orientation, flipping one board relative to the other would mirror the X and Y field components reported by that board, causing the center-field average (Equation 1) to cancel real signal rather

than noise. Verify that the silkscreen text ("BOILER UP!!") is readable from the same side on both boards when installed.

4. For two-board operation, mount Board 0 and Board 1 at different axial positions along the cavity (or at 45° angular offset) to sample additional field locations.
5. Connect the FFC cable between Board 0 PIN_OUT and Board 1 PIN_IN *before* routing the assembly into the chamber.

4.3 Cable Routing and Vacuum Feedthrough

The electrical connection between the sensor boards (inside the chamber) and the Arduino control box (outside) is made via the FFC cable and the Arduino USB cable. These must be routed through the vacuum chamber wall:

- Keep the FFC cable length under 30 cm inside the chamber where possible. Longer cables increase bus capacitance and may require reducing the I^2C clock to 100 kHz (`Wire.setClock(100000)`) to maintain reliable communication.
- The Arduino Nano Every and its USB cable operate at room temperature and ambient pressure outside the chamber. No special provisions are needed for the external components.

4.4 Control Box and Computer Connection

The Arduino Nano Every connects to the data acquisition computer via a standard USB-A to USB-Micro cable. On Windows it appears as a COM port; on Linux/macOS as `/dev/ttyACM0` (or similar). The Python logger script (`python/logger.py` in the repository) opens this port at 115200 baud and records timestamped CSV data automatically.

5 Two-Board Setup

5.1 Overview

Two identical boards are connected in series via FFC cable to provide eight sensor channels. Both boards share the same SDA, SCL, RST, and power lines. The two TCA9546A multiplexers are differentiated by I^2C address:

- **Board 0:** Mux address `0x70` (A0 = GND, factory default)
- **Board 1:** Mux address `0x71` (A0 = VCC, requires rework)

5.2 Hardware Modification for Board 1

Both boards are fabricated from *identical* Gerber files — a deliberate choice to simplify manufacturing (one design, two copies ordered). The trade-off is that both muxes ship with A0 tied to GND (`0x70`). When two boards share the same I^2C bus this creates an address conflict, making Board 1 invisible to the Arduino. One board must be reworked to assign it `0x71`.

A future PCB revision should add a 3-pad solder jumper on the A0 net to allow address selection without rework.

To perform the rework on Board 1's TCA9546A (U1):

1. Identify the A0 pad (pin 1 of TCA9546AD; see Figure 4).
2. Remove or cut the GND connection to A0.
3. Bridge A0 to the VCC net (3.3 V) with a short wire or solder bridge.
4. Confirm with an ohmmeter that A0 is connected to VCC and isolated from GND.

5.3 FFC Cable Connection

1. Use a **Type-B** FFC cable (Top-on-One-Side, Bottom-on-the-Other). This type is required because PIN_OUT on one board faces opposite to PIN_IN on the next.
2. Insert one end into Board 0's PIN_OUT (J1); insert the other end into Board 1's PIN_IN (J4).
3. **Before powering on:** verify that +5 V, +3.3 V, GND, SDA, and SCL map correctly across the connector. A reversal will short power to GND.

5.4 Sensor Numbering Convention

Table 10: Full 8-Sensor Addressing Map

Logical ID	Board	Mux addr	Channel	Sensor addr
Sensor 0	Board 0	0x70	0 (0x01)	0x30
Sensor 1	Board 0	0x70	1 (0x02)	0x30
Sensor 2	Board 0	0x70	2 (0x04)	0x30
Sensor 3	Board 0	0x70	3 (0x08)	0x30
Sensor 4	Board 1	0x71	0 (0x01)	0x30
Sensor 5	Board 1	0x71	1 (0x02)	0x30
Sensor 6	Board 1	0x71	2 (0x04)	0x30
Sensor 7	Board 1	0x71	3 (0x08)	0x30

6 Firmware and Data Acquisition

6.1 Development Environment

- Arduino IDE 2.x
- Board package: *Arduino megaAVR Boards* (Boards Manager → search "megaAVR")
- Board: **Arduino Nano Every**; Registers Emulation: **None (ATMEGA4809)**
- Required libraries: `Wire` (built-in), *[MMC5983MA library — add name and source]*
- Baud rate: 115200

6.2 Firmware Algorithm

The firmware runs a continuous acquisition loop on the Arduino Nano Every. Full source code is provided in Appendix D; the algorithm is described here.

Initialization (runs once): The I^2C bus is configured at 400 kHz (Fast Mode). The RST pin is held high (deasserted) and the mux RESET line is pulsed briefly to clear any stuck bus state. A SET pulse is then issued to all eight sensors in sequence, cycling through each mux channel on both boards. This establishes the initial magnetization reference. The firmware also runs an I^2C address scan and prints results over serial — useful for hardware bring-up verification.

Main loop (repeats at ∼10 Hz): For each of the eight sensors in order (Board 0 channels 0–3, then Board 1 channels 0–3):

1. Write the channel select byte to the appropriate mux (`0x70` or `0x71`) to connect that sensor.
2. Trigger a single measurement by writing `0x01` to the sensor's `CONTROL_0` register.
3. Poll the `STATUS` register until the Meas_M_Done bit (bit 0) is set, indicating conversion is complete (∼2 ms).
4. Burst-read registers `0x00`–`0x06` and assemble three 18-bit integers r_x, r_y, r_z. The two LSBs of each axis are packed into register `0x06`.
5. Convert raw counts to Gauss using Equation 3.
6. Write `0x00` to the mux to disconnect the channel.
7. Transmit the result over USB-serial in CSV format.

6.3 Raw-to-Field Conversion

The MMC5983MA outputs 18-bit unsigned integers. Zero field corresponds to the midscale code $2^{17} = 131072$; the full-scale range of ±8 G spans $2^{18} = 262144$ counts. The conversion

is:

$$B\,[\mathrm{G}] = \frac{r - 2^{17}}{2^{18}/16} = \frac{r - 131072}{16384} \tag{3}$$

where r is the 18-bit raw output value. At 0.0625 mG per LSB, this gives a theoretical noise floor well below the residual fields typically found in a lab environment.

6.4 Center-Field Average

After converting all sensor readings to Gauss, the firmware computes the spatial average over all N sensors per board (Equation 1). For a single board with $N = 4$:

$$\bar{B}_x = \frac{B_{x,0} + B_{x,1} + B_{x,2} + B_{x,3}}{4}, \quad \bar{B}_y = \frac{B_{y,0} + \cdots + B_{y,3}}{4}, \quad \bar{B}_z = \frac{B_{z,0} + \cdots + B_{z,3}}{4} \tag{4}$$

This average is also transmitted over serial and is the primary quantity used to monitor and minimize the magnetic field at the experiment location.

6.5 SET/RESET Offset Cancellation

For improved accuracy, especially over long runs, the optional SET/RESET subtraction procedure (Equation 2) can be performed periodically. In the firmware, enabling `USE_OFFSET_CANCEL = true` activates this mode: a SET measurement and a RESET measurement are taken on every reading cycle and the difference divided by two. This cancels the temperature-dependent bridge offset at the cost of doubling the per-sensor acquisition time. For routine monitoring a single SET at startup followed by triggered measurements is generally sufficient.

Issue a manual SET/RESET at the start of each acquisition session and whenever readings drift unexpectedly (recommended interval: every 5–10 minutes during long runs).

6.6 Data Collection Workflow (for Chunya)

1. Connect the Arduino Nano Every to the computer via USB. On Windows select the correct COM port; on Linux/macOS use `/dev/ttyACM0`.
2. Run the Python logger from the repository: `python python/logger.py --port COM3 --output data.csv`
3. Allow 30 seconds of warm-up before recording; the logger handles this automatically unless `--no-warmup` is specified.
4. The logger saves timestamped CSV with columns `time_s, sensorID, Bx_G, By_G, Bz_G` and prints a live per-sensor summary including field magnitude and noise floor.
5. To stop: press Ctrl-C. The logger prints a per-sensor summary and flags any sensors with unexpectedly low or zero readings.
6. Verify that all 8 sensor IDs appear in the data and that field magnitudes are in the range 0.2–0.6 G (ambient Earth field).

7 Troubleshooting

Table 11: Common Issues and Resolutions

Symptom	Likely Cause	Resolution
I^2C scanner shows nothing	Pull-ups missing; bad solder joint on mux	Verify 2.7 kΩ on SDA/SCL; reflow U1
Only one mux found (e.g., `0x70` only)	Board 1 A0 not moved to VCC	Rework A0 on Board 1 TCA9546A (Section 5)
Mux found but `0x30` missing on a channel	Bad reflow on MMC5983MA; wrong channel byte	Re-solder suspect sensor; check channel select logic
All-zeros or maxed-out readings	SET not issued; SPL_CS not floating (SPI mode)	Issue SET before measurement; confirm SPL_CS is NC
Noisy / drifting readings	Insufficient SET/RESET; ferromagnetic object nearby	Repeat SET/RESET; remove tools/cables from sensor area
FFC cable connects but no comms	Type-A cable used (wrong orientation)	Replace with Type-B FFC cable
Upload fails (Nano Every)	Wrong Registers Emulation setting	Set to "None (ATMEGA4809)"; double-tap reset
LM3940 output ≠ 3.3 V	C2 wrong polarity; U2 mis-oriented; ceramic cap used	Check C2 polarity; confirm C2 is tantalum, not ceramic

7.1 Using an Oscilloscope to Debug I^2C

Probe SDA (U1 pin 15) and SCL (U1 pin 14). A healthy 400 kHz transaction should show: SCL as a clean square wave at ≈3.3 V logic high; SDA transitioning only while SCL is low; and SDA pulled low by the device on the 9th (ACK) clock pulse. If SDA does not return high between transactions, a device is holding the bus low. Assert RST_PIN low for $> 10\,\mu$s to reset the mux and clear the stuck state.

7.2 Bus Recovery Procedure

If the I^2C bus hangs (SDA stuck low): (1) assert RST_PIN low from the Arduino for ∼100 μs; (2) release high; (3) call `Wire.end()` then `Wire.begin()` to reinitialize the TWI peripheral; (4) re-run the I^2C scanner to confirm recovery.

8 Known Limitations and Future Work

- **No address-select solder jumper:** Both boards are fabricated from identical Gerbers, requiring a manual rework on Board 1 to change the TCA9546A address. A future revision should add a 3-pad solder jumper on the A0 net.
- **No on-chip temperature compensation:** MMC5983MA offset drifts with temperature. Repeat SET/RESET periodically for long runs.
- **No interrupt-driven acquisition:** The INT pin is unused; firmware polls the Meas_M_Done bit. Wiring INT to the Arduino in a future revision would reduce acquisition latency and CPU load.
- **Expansion limited to 8 boards:** The TCA9546A address range (`0x70`–`0x77`) supports at most 8 boards (32 sensors) on one I^2C bus.
- **FFC cable length sensitivity:** Cables >30 cm inside the vacuum chamber increase bus capacitance. Reduce I^2C to 100 kHz if communication errors occur.

Acknowledgments

Thank you to Dr. Elliott for guidance throughout this project, and to Chunya Wang for testing the two-board configuration, identifying the I^2C address conflict (and its solution) during bring-up, and providing context on the experimental requirements.

References

[1] MEMSIC, *MMC5983MA 3-Axis Magnetic Sensor with I²C/SPI Interface Datasheet*, Rev. A.

[2] Texas Instruments, *TCA9546A Low-Voltage 4-Channel I²C Switch with Reset Datasheet*, SCPS204.

[3] Texas Instruments, *LM3940 1-A Low-Dropout Regulator for 5-V to 3.3-V Conversion Datasheet*, SNOS386.

[4] Arduino, *Arduino Nano Every Datasheet and Pinout Reference*, https://docs.arduino.cc/hardware/nano-every.

[5] KiCad EDA, https://www.kicad.org.

A I²C Address Map

Table 12: Complete I²C Address Map (two-board system)

Device	Address	Board	Notes
TCA9546A (U1)	0x70	Board 0	A0 = GND
TCA9546A (U1)	0x71	Board 1	A0 = VCC (rework required)
MMC5983MA (U3–U6)	0x30	Both	Isolated per mux channel
Arduino Nano Every	—	—	Master only

B Register Quick Reference

Table 13: MMC5983MA Key Registers

Reg (hex)	Name	R/W	Description
0x00	Xout0	R	X-axis output [17:10]
0x01	Xout1	R	X-axis output [9:2]
0x02	Yout0	R	Y-axis output [17:10]
0x03	Yout1	R	Y-axis output [9:2]
0x04	Zout0	R	Z-axis output [17:10]
0x05	Zout1	R	Z-axis output [9:2]
0x06	XYZout2	R	Bits [1:0] of X, Y, Z (18-bit extension)
0x08	Status	R	Bit 0: Meas_M_Done
0x09	Control_0	W	Bit 0: TM_M (trigger); Bit 3: SET; Bit 4: RESET
0x0A	Control_1	W	BW[1:0]: bandwidth / ODR select
0x2F	Product ID	R	Should read `0x30`

C Gerber File Manifest

All Gerber files were generated from KiCad 10.0.1 on 2026-04-27. Submit all files plus the `.gbrjob` to the PCB fabricator; the job file encodes board size, layer count, thickness, and design rules.

Table 14: Gerber File Manifest

Filename	Layer	Function
`MAG PCB FINAL-F_Cu.gbr`	F_Cu (L1)	Top copper, all signal routing
`MAG PCB FINAL-B_Cu.gbr`	B_Cu (L2)	Bottom copper, return paths
`MAG PCB FINAL-F_Paste.gbr`	F_Paste	Top solder paste (SMT stencil)
`MAG PCB FINAL-B_Paste.gbr`	B_Paste	Bottom solder paste
`MAG PCB FINAL-F_Silkscreen.gbr`	F_Silkscreen	Top component labels
`MAG PCB FINAL-B_Silkscreen.gbr`	B_Silkscreen	Bottom labels
`MAG PCB FINAL-F_Mask.gbr`	F_Mask	Top solder mask openings
`MAG PCB FINAL-B_Mask.gbr`	B_Mask	Bottom solder mask openings
`MAG PCB FINAL-Edge_Cuts.gbr`	Edge_Cuts	Board outline / routed profile
`MAG PCB FINAL-job.gbrjob`	—	Gerber job file

D Firmware Source Code

I²C Scanner (bring-up verification)

```
#include <Wire.h>
void setup() {
  Wire.begin(); Wire.setClock(400000); Serial.begin(115200);
}
void loop() {
  for (uint8_t addr = 1; addr < 127; addr++) {
    Wire.beginTransmission(addr);
    if (Wire.endTransmission() == 0) {
      Serial.print("Found: 0x"); Serial.println(addr, HEX);
    }
  }
  Serial.println("-- scan complete --"); delay(3000);
}
```

Listing 1: I²C scanner sketch

Expected output (two boards):

```
Found: 0x70   <- Board 0 mux
Found: 0x71   <- Board 1 mux
```

Main Acquisition Firmware

```
#include <Wire.h>

#define MUX0     0x70
#define MUX1     0x71
#define MMC_ADDR 0x30
#define RST_PIN  4
#define ZERO     131072UL
#define SCALE    16384.0f   // counts per Gauss

void selectCh(uint8_t mux, uint8_t ch) {
  Wire.beginTransmission(mux); Wire.write(1<<ch); Wire.
     endTransmission(); }
void deselect(uint8_t mux) {
  Wire.beginTransmission(mux); Wire.write(0x00); Wire.
     endTransmission(); }

bool readSensor(float &Bx, float &By, float &Bz) {
  Wire.beginTransmission(MMC_ADDR);
  Wire.write(0x09); Wire.write(0x01);   // CONTROL_0: trigger
     measurement
  Wire.endTransmission();
```

```
  uint32_t t0=millis();
  while (true) {                              // poll Meas_M_Done
    Wire.beginTransmission(MMC_ADDR); Wire.write(0x08); Wire.
       endTransmission(false);
    Wire.requestFrom((uint8_t)MMC_ADDR,(uint8_t)1);
    if (Wire.available() && (Wire.read()&0x01)) break;
    if (millis()-t0 > 10) return false;
  }
  Wire.beginTransmission(MMC_ADDR); Wire.write(0x00); Wire.
     endTransmission(false);
  Wire.requestFrom((uint8_t)MMC_ADDR,(uint8_t)7);
  uint8_t b[7]; for(int i=0;i<7&&Wire.available();i++) b[i]=Wire.
     read();
  uint32_t rx=((uint32_t)b[0]<<10)|((uint32_t)b[1]<<2)|((b[6]>>6)&3)
     ;
  uint32_t ry=((uint32_t)b[2]<<10)|((uint32_t)b[3]<<2)|((b[6]>>4)&3)
     ;
  uint32_t rz=((uint32_t)b[4]<<10)|((uint32_t)b[5]<<2)|((b[6]>>2)&3)
     ;
  Bx=((float)rx-ZERO)/SCALE;
  By=((float)ry-ZERO)/SCALE;
  Bz=((float)rz-ZERO)/SCALE;
  return true;
}

void doSet(uint8_t mux, uint8_t ch) {
  selectCh(mux,ch);
  Wire.beginTransmission(MMC_ADDR); Wire.write(0x09); Wire.write(0
     x08);
  Wire.endTransmission(); delayMicroseconds(1); deselect(mux); }

void setup() {
  Wire.begin(); Wire.setClock(400000); Serial.begin(115200);
  pinMode(RST_PIN,OUTPUT); digitalWrite(RST_PIN,HIGH);
  // Issue SET on all 8 sensors at startup
  for(uint8_t ch=0;ch<4;ch++) { doSet(MUX0,ch); doSet(MUX1,ch); }
  Serial.println("sensorID,Bx_G,By_G,Bz_G,avg_Bx,avg_By,avg_Bz");
}

void loop() {
  float Bx,By,Bz;
  float sumX=0,sumY=0,sumZ=0; uint8_t n=0;

  // Board 0 (sensors 0-3)
  for(uint8_t ch=0;ch<4;ch++) {
    selectCh(MUX0,ch);
    if(readSensor(Bx,By,Bz)) {
```

```
      Serial.print(ch);Serial.print(',');
      Serial.print(Bx,4);Serial.print(',');
      Serial.print(By,4);Serial.print(',');
      Serial.println(Bz,4);
      sumX+=Bx; sumY+=By; sumZ+=Bz; n++;
    }
    deselect(MUX0);
  }
  // Board 1 (sensors 4-7)
  for(uint8_t ch=0;ch<4;ch++) {
    selectCh(MUX1,ch);
    if(readSensor(Bx,By,Bz)) {
      Serial.print(ch+4);Serial.print(',');
      Serial.print(Bx,4);Serial.print(',');
      Serial.print(By,4);Serial.print(',');
      Serial.println(Bz,4);
      sumX+=Bx; sumY+=By; sumZ+=Bz; n++;
    }
    deselect(MUX1);
  }
  // Print center-point average (Eq. 1)
  if(n>0) {
    Serial.print("AVG,");
    Serial.print(sumX/n,4);Serial.print(',');
    Serial.print(sumY/n,4);Serial.print(',');
    Serial.println(sumZ/n,4);
  }
  delay(100);
}
```

Listing 2: Main firmware (magnetometer.ino)